ARAYASELASSIE ABEBE SEMU, PhD

ENVIRONMENTAL SCIENTIST | ECOLOGIST | BIODIVERSITY & GIS SPECIALIST | PROJECT LEAD

Cork / Skibbereen, Ireland | +353 85 759 6544 | pbbmaraya@gmail.com | Stamp 4

Portfolio website: https://magnificent-biscotti-ed1748.netlify.app/

Professional Profile

PhD-qualified environmental scientist and ecologist with 10+ years of postgraduate experience across forest and wetland ecology, botany, biodiversity assessment, restoration, GIS/remote sensing, quantitative analysis, technical reporting and environmental project delivery. Recently completed a Government of Ireland postdoctoral fellowship at University College Cork, leading a €119,000 Research Ireland-funded programme across 34 relict forest sites. Combines strong field ecology with R/GIS, data QA/QC, project leadership, budgets, stakeholder engagement and peer-reviewed scientific communication. Experienced with ArcGIS, QGIS, Google Earth Engine, GIS in R, MaxEnt, Git-supported research workflows, taxonomy validation using World Flora Online and Darwin Core concepts. Seeking ecology, environmental science, biodiversity, restoration, GIS/data or environmental project roles in Ireland.

Core Strengths

Ecology & biodiversity	Botany, vegetation surveys, forest/wetland ecology, habitat assessment, regeneration, restoration, biodiversity baselines.
GIS & data	R, ArcGIS, QGIS, Google Earth Engine, GIS in R, remote sensing, MaxEnt, spatial QA/QC, Git, Darwin Core familiarity.
Project delivery	Multi-site coordination, workplans, budgets, procurement, QA/QC, reporting, stakeholder liaison and team leadership.
Communication	Technical reports, publications, presentations, teaching, mentoring and non-specialist communication.

Professional Experience

Government of Ireland Postdoctoral Fellow / Postdoctoral Researcher | University College Cork | Sep 2024-Aug 2026

- Led a €119,000 Research Ireland-funded ecology programme across 34 relict forest sites, coordinating botanical fieldwork, GIS/R analysis, schedules, budgets, QA/QC, collaborators and reporting.
- Analysed vegetation composition, forest structure, regeneration, disturbance, fragmentation and connectivity using field data, ArcGIS, Google Earth Engine, GIS in R and R.
- Built reproducible workflows for ecological mapping, spatial analysis and research-data management; used Git for data/analytical materials associated with recent publications.
- Prepared peer-reviewed manuscripts, technical outputs, maps and presentations; mentored students in plant identification, survey design, data quality and analysis.

Volunteer Field Ecologist & Steering Committee Member | Actions for Nature Skibbereen / CECAS | Jan 2024-Present

- Support Irish habitat identification and mapping using Fossitt classification, field observations, GIS and biodiversity planning.
- Contribute ecological evidence, steering-group input and accessible communication for community conservation and habitat-enhancement work.

Lecturer, Researcher, Project Lead & Former Head of Department | Haramaya University | Nov 2015-Jan 2024

- Led forest, wetland, protected-area, restoration and urban-ecology projects from study design and field sampling through analysis, reporting and publication.
- Conducted botanical surveys, vegetation structure/regeneration assessment, biodiversity baselines, ecosystem-service studies and climate-change habitat-suitability modelling.
- Managed teams, budgets, procurement, grant documentation, project reporting and academic programmes; supervised undergraduate and postgraduate research.
- Applied R-based multivariate ecology, diversity metrics and GIS; used authoritative taxonomic sources to validate and harmonise plant names.

Field Ecologist / Biodiversity Consultant | Chebera Churchura National Park | Project-based, 2022

- Delivered biodiversity and habitat-condition baseline assessment, identifying ecological hotspots, pressures and conservation priorities.
- Prepared restoration, monitoring and management recommendations for protected-area planning.

Selected Projects

- Relict Forest Patch Ecology & Connectivity, Ireland - 34-site forest ecology, regeneration, fragmentation, connectivity and GIS/R programme.
- Northern Wollo Monastery Forests - plant diversity, vegetation structure, regeneration, ecosystem services and restoration research.
- Haramaya, Teneke & Adele Wetlands - ecological assessment, monitoring, project leadership and management recommendations.
- Babile Elephant Sanctuary - habitat fragmentation, connectivity, restoration suitability and conservation-planning assessment.

Education

- PhD, Plant Biology & Biodiversity Management - Addis Ababa University, 2023
- MSc, Plant Biology & Biodiversity Management - Addis Ababa University, 2013
- BEd, Biology - Bahir Dar University, 2011
- BA, Economics - Bahir Dar University

Technical Skills

R (vegan, ggplot2, sf, terra/raster/sp); ArcGIS; QGIS; Google Earth Engine; GIS in R; remote sensing; GPS; MaxEnt; multivariate community analysis; diversity metrics; spatial analysis; data cleaning/validation; QA/QC; Git; Darwin Core familiarity; World Flora Online taxonomy workflows; Microsoft Excel, Word, PowerPoint and Outlook.

Professional Development & Memberships

- IEMA Foundation Certificate in Environmental Management; ISO 14001:2015 Environmental Management Systems training; Principles of Environmental Impact Assessment.
- GIS in R and Google Earth Engine; National Forest Survey / National Forest Inventory; environmental monitoring and aquatic pollution.
- CIEEM membership application submitted and currently under review; professional affiliations include British Ecological Society and International Association for Vegetation Science.

Selected Publications

- Abebe, A., Eichhorn, M.P. & Crego, R.D. (2026). Spatiotemporal dynamics and cross-boundary connectivity of an ecological network across a fragmented landscape. Landscape Ecology.
- Semu, A.A. & Abegaz, A.W. (2026). Intraspecific genetic diversity as a driver of ecosystem functioning. Frontiers in Ecology and Evolution, 14, 1862503.
- Abebe, A., Bekele, T. & Lulekal, E. (2025). Plant species diversity, vegetation structure and regeneration of Northern Wollo Monastery Forests. PLOS ONE, 20(9), e0330689.

Additional Information

- Stamp 4 - unrestricted right to work in Ireland; no sponsorship required.
- Based in Cork / Skibbereen; open to roles across Ireland and hybrid arrangements.
- Own car; Irish learner permit; driving test rescheduled for late December 2026.
- Portfolio: https://magnificent-biscotti-ed1748.netlify.app/
- References available on request.